41.4: Evolutionary adaptations of vertebrate digestive systems correlate with diet

EVOLUTION The digestive systems of vertebrates are varied, with many adaptations linked to an animal's diet. To highlight how form fits function, we'll examine a few of them.

Dental Adaptations

Dentition, an animal's assortment of teeth, is one example of structural variation reflecting diet **(Figure 41.15)**. The evolutionary adaptation of teeth for processing different kinds of food is one of the major reasons that mammals have been so successful. For example, a sea otter uses its sharp canine teeth to tear apart prey such as crabs and its slightly rounded molars to crush their shells. Nonmammalian vertebrates generally have less specialized dentition, but there are interesting exceptions. Venomous snakes, such as rattlesnakes, have fangs, modified teeth that inject venom into prey. Some fangs are hollow, like syringes, whereas others drip the toxin along grooves on the surfaces of the teeth.

Figure 41.15

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Dentition and diet.
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Arrangement of incisors, canines, premolars and molars in carnivores, herbivores and omnivores. Dentition patterns correspond to diet type.

Arrangement of incisors, canines, premolars and molars in carnivores, herbivores and omnivores. Teeth are described from front to back.

Carnivore. Carnivores, such as members of the dog and cat families, generally have large, pointed incisors; canines are prominent and also pointed. Both can be used to kill prey and rip or cut away pieces of flesh. The jagged premolars and molars cut and shred food.

Herbivore. Herbivores, such as horses and deer, generally have incisors and canines that are modified for biting off pieces of vegetation. In the skull shown there is a gap in the elongated jaw between the canines and premolars. Herbivores usually have premolars and molars with broad, ridged surfaces that grind tough plant material.

Omnivore. As omnivores, humans are adapted to eating both plants and meat. Adults have 32 teeth. From front to back along either side of the mouth are four blade-like incisors for biting, a pair of pointed canines

for tearing, four premolars for grinding, and six molars with a notched top surface for crushing.

Stomach and Intestinal Adaptations

Evolutionary adaptations to differences in diet are sometimes apparent as variations in the dimensions of digestive organs. For example, large, expandable stomachs are common in carnivorous vertebrates, which may wait a long time between meals and must eat as much as they can when they do catch prey. An expandable stomach enables a rock python to ingest a whole gazelle (see Figure 41.5) and a 200-kg African lion to consume 40kg of meat in one meal!

Adaptation is also apparent in the length of the digestive system in different vertebrates. In general, herbivores and omnivores have longer alimentary canals relative to their body size than do carnivores. Plant matter is more difficult to digest than meat because it contains cell walls. A longer digestive tract furnishes more time for digestion and more surface area for nutrient absorption. As an example, consider the lion and koala in **Figure 41.16**. The koala has much longer intestines, relative to its size, enhancing the processing of fibrous, protein-poor eucalyptus leaves from which it obtains nearly all of its nutrients and water.

Figure 41.16

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    The alimentary canals of a carnivore (lion) and herbivore
(koala).
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Comparison of alimentary canals of a coyote and koala. The alimentary canal of the koala is much longer, with a greatly increased cecum and colon as compared to the coyote.

The relatively short digestive tract of the lion is sufficient for digesting meat and absorbing its nutrients. In contrast, the koala's long alimentary canal is specialized for digesting eucalyptus leaves. Extensive chewing chops the leaves into tiny pieces, increasing exposure to digestive juices. In the long cecum and the upper portion of the colon, symbiotic bacteria further digest the shredded leaves, releasing nutrients that the koala can absorb.

Comparison of the alimentary canals of a coyote (carnivore) and koala (herbivore). The stomachs of both animals are similar. Following the stomach is the small intestine, and then the cecum for both animals. The coyote cecum is a small intestinal extension, but the koala cecum is a long, coiled loop. Likewise, the colon (large intestine) of the coyote is relatively short as compared to the koala, which has a much longer colon.

Mutualistic Adaptations

An estimated 10–100 trillion bacteria live in the human digestive system. The coexistence of humans and many intestinal bacteria is an example of mutualism, an interaction between two species that benefits both of them (see Concept 54.1). For example, some intestinal bacteria produce vitamins, such as vitamin K, biotin, and folic acid, which are absorbed into the blood, supplementing our dietary intake. Intestinal bacteria also regulate the development of the intestinal epithelium and the function of the innate immune system. The bacteria in turn receive a steady supply of nutrients and a stable host environment.

Recently, we have greatly expanded our knowledge of the **microbiome**, the collection of microorganisms living in and on the body, along with their genetic material. To study the microbiome, scientists are using a DNA sequencing approach based on the polymerase chain reaction (PCR, see Figure 20.8). To date they have found more than 400 bacterial species in the human digestive tract, a far greater number than had been identified through approaches relying on laboratory culture and characterization. Furthermore, researchers have found significant differences in the microbiome associated with diet, disease, and age **(Figure 41.17)**.

Figure 41.17

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Variation in human gut microbiome at different life stages.
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Comparison of the relative proportions of firmicutes, bacteroidetes, actinobacteria, proteobacteria, and other bacterial phyla in human infants, toddlers, adults, and elderly people.

By copying and sequencing bacterial DNA in samples obtained from intestinal tracts of humans of different ages, researchers characterized the bacterial community that makes up the human gut and how it changes with age.

In a human infant on breastmilk, the gut microbiome consists of about 50% actinobacteria, 25% proteobacteria, 13% firmicutes, 10% bacteroidetes, and 2% other bacterial phyla. In a human infant on formula, the gut microbiome consists of about 45% actinobacteria, 45% firmicutes, and 10% proteobacteria. In a human infant on solid food, the gut microbiome consists of about 45% bacteroidetes, 33% firmicutes, 20% other bacterial phyla, and 2% proteobacteria. In a human toddler who is healthy, the gut microbiome consists of about 45% bacteroidetes, 33% firmicutes, 20% other bacterial phyla, and 2% proteobacteria. In a human toddler on antibiotic treatment, the gut microbiome consists of about 78% bacteroidetes, 21% firmicutes, and 1% proteobacteria. In a human toddler who is malnourished, the gut microbiome consists of about 40% proteobacteria, 33% firmicutes, 20% bacteroidetes, 5% other bacterial phyla, and 2% actinobacteria. In a healthy human adult, the gut microbiome consists of about 75% firmicutes, 19% bacteroidetes, 3% actinobacteria, 2% proteobacteria, and 1% other bacterial phyla. In an obese human adult, the gut microbiome consists of about 65% firmicutes, 15% bacteroidetes, 12% proteobacteria, and 8%

actinobacteria. In an elderly human aged 65-80 years old, the gut microbiome consists of about 78% firmicutes, 14% bacteroidetes, 4% actinobacteria, 3% proteobacteria, and 1% other bacterial phyla. In an elderly human over 100 years old, the gut microbiome consists of about 60% firmicutes, 20% actinobacteria, 18% bacteroidetes, 1% proteobacteria, and 1% other bacterial phyla. All percentages are approximate.

INTERPRET THE DATA *Using data displayed in**Figures 41.17**and*41.18*, compare the relative abundance of Actinobacteria in the microbiome of a healthy adult's intestinal tract to that in a healthy stomach. Suggest a possible explanation for why the microbiome composition in the two organs is different even though the intestine and stomach are directly connected.*

One example of the power of microbiome studies comes from research on gastric ulcers, a disease that damages the stomach lining. Experiments demonstrating that ulcers are caused by infection with the acid-tolerant bacterium *Helicobacter pylori* and can be cured with antibiotics earned Australian researchers Barry Marshall and Robin Warren a Nobel Prize in 2005.

Recently, scientists analyzed the microbiome in samples from human stomachs to learn how *H. pylori* infection leads to ulcers. Their findings were dramatic: *H. pylori* infection led to the near-complete elimination of all other bacterial species commonly found in the stomach **(Figure 41.18)**.

Figure 41.18

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The stomach microbiome and stomach health.
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Bar chart of percent of sampled stomach bacteria by bacterial phylum in individuals uninfected and infected with H. pylori. When infected, H. pylori makes up over 95% of bacteria present.

Bar chart of percent of sampled stomach bacteria by bacterial phylum in individuals uninfected and infected with H. pylori. The horizontal axis is bacterial phylum with a bar for each phylum, and the vertical axis is percent of sampled stomach bacteria, ranging from 0 to 100 with increments every 20 percent. Uninfected individuals have bacteria present from the phyla (in decreasing order of percent sampled) Actinobacteria, Firmicutes, Bacteroidetes, Proteobacteria and Fuscobacteria; infected individuals have more than 95% H. pylori and only negligible occurrences of other phyla. H. pylori are rod-shaped bacteria with flagella.

In samples from individuals infected with *Helicobacter pylori*, more than 95% of the sequences were from that species, which belongs to the phylum Proteobacteria. The stomach microbiome in uninfected individuals was much more diverse.

Studies on the microbiome have already led to therapies for intestinal infections by antibiotic-resistant pathogens, a major public health problem. One such therapy is fecal microbial transplantation, in which the microbiome of a healthy individual is introduced into the patient's intestine. This approach has been used

for otherwise untreatable diarrhea caused by the bacterium *Clostridium difficile* (**Figure 41.19)**. Such infections are most common after antibiotic treatment has killed off the resident microbiome. Clinical trials have shown the fecal transplantation approach to be effective, although there is a real risk of secondary infection.

Figure 41.19

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*Clostridium difficile.*
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A bacterial cell with many flagella.

Another treatment against antibiotic-resistant pathogens makes use of bacteriophages, viruses that infect bacteria but not human cells (see Concept 19.1). In 2017, bacteriophages engineered to kill a multidrug-resistant bacterial pathogen, *Acinetobacter baumannii*, were successfully used to restore health to a severely affected patient.

Interview

Interview with Steffanie Strathdee: Harnessing phages to fight deadly infections

Steffanie Strathdee.

Mutualistic Adaptations in Herbivores

Mutualistic relationships with microorganisms are also very important in herbivores. Herbivores get much of the chemical energy they need from the cellulose of plant cell walls, but, like other animals, herbivores do not produce enzymes that hydrolyze cellulose. Instead, many vertebrates (as well as termites, whose wooden diets consist largely of cellulose) host large populations of mutualistic bacteria and protists in fermentation chambers in their alimentary canals. These symbiotic microorganisms have enzymes that can digest cellulose to simple sugars and fatty acids that the animal can absorb. In many cases, the microorganisms also use the sugars from digested cellulose in the production of a variety of essential nutrients, such as vitamins and amino acids.

In horses, koalas, and elephants, symbiotic microorganisms are housed in a large cecum. In contrast, the hoatzin, an herbivorous bird found in South American rain forests, hosts microorganisms in a large, muscular crop (an esophageal pouch; see Figure 41.7). Hard ridges in the wall of the crop grind plant leaves into small fragments, and the microorganisms break down cellulose.

In rabbits and some rodents, mutualistic bacteria live in the large intestine as well as the cecum. Since most nutrients are absorbed in the small intestine, nourishing by-products of fermentation by bacteria in the large intestine are initially lost with the feces. Rabbits and rodents recover these nutrients by *coprophagy* (from the Greek, meaning "dung eating"), feeding on some of their feces and then passing the food through the alimentary canal a second time. The familiar rabbit "pellets," which are not reingested, are the feces eliminated after food has passed through the digestive tract twice.

The most elaborate adaptations for an herbivorous diet have evolved in the animals called ***ruminants***, the cud-chewing animals that include deer, sheep, and cattle **(Figure 41.20)**.

Figure 41.20

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Ruminant digestion.
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Path of food (grass) through cow is mouth, esophagus, rumen, reticulum (with regurgitation to mouth for re-chewing cud), omasum, abomasum, intestine.

The stomach of a cow, a ruminant, has four chambers. ***1*** *Chewed food first enters the rumen and reticulum, where mutualistic microorganisms digest cellulose in the plant material.* ***2*** *Periodically, the cow regurgitates and rechews "cud" from the reticulum, further breaking down fibers and thereby enhancing microbial action.* ***3*** *The reswallowed cud passes to the omasum, where some water is removed.* ***4*** *It then passes to the abomasum for digestion by the cow's enzymes. In this way, the cow obtains significant nutrients from both the grass and the mutualistic microorganisms, which maintain a stable population in the rumen.*

Although we have focused our discussion on vertebrates, adaptations related to digestion are also widespread among other animals. Some of the most remarkable examples are the over 3-meter-long giant tubeworms **(Figure 41.21)** that live at pressures as high as 260 atmospheres around deep-sea hydrothermal vents (see Figure 52.16). These worms have no mouth or digestive system. Instead, they obtain all of their energy and nutrients from mutualistic bacteria that live within their bodies. The bacteria carry out chemoautotrophy (see Concept 27.3) using the carbon dioxide, oxygen, hydrogen sulfide, and nitrate available at the vents. Thus, for invertebrates and vertebrates alike, the evolution of mutualistic relationships with symbiotic microorganisms is an adaptation that expands the sources of nutrition available to animals.

Figure 41.21

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Giant tubeworm, an animal without a digestive system.
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Giant tubeworm with highly vascularized red plume at free end.

Having examined how animals optimize their extraction of nutrients from food, we'll next turn to the challenge of balancing the use of these nutrients.

Concept 41.5 Feedback circuits regulate digestion, energy storage, and appetite

To complete our consideration of animal nutrition, we'll explore the ways that obtaining and using nutrients are matched to an animal's circumstances and need for energy.

Regulation of Digestion

Many animals face long gaps between meals. Under such circumstances, there is no need for their digestive systems to be active continuously. Instead, processing is activated stepwise. As food reaches each new compartment, it triggers the secretion of digestive juices for the next stage of processing. Muscular contractions then move the contents farther along the canal. For example, you learned earlier that nervous reflexes stimulate the release of saliva when food enters the oral cavity and orchestrate swallowing when a bolus of food reaches the pharynx. Similarly, the arrival of food in the stomach triggers churning and the release of gastric juices. These events, as well as movement through the small and large intestines, are regulated by the *enteric nervous system*, a network of neurons dedicated to the digestive organs.

The endocrine system also plays a critical role in controlling digestion. As described in **Figure 41.22**, a series of hormones released by the stomach and duodenum help ensure that digestive secretions are present only when needed. Like all hormones, they are transported through the bloodstream. This is true even for the hormone gastrin, which is secreted by the stomach and targets that same organ.

Figure 41.22

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Hormonal control of digestion.
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Inhibition and stimulation highlighted in the hormonal control of digestion, which involves the stomach, gall bladder and pancreas, as well as various hormones.

Steps are as follows.

1. Diagram of large liver connected to small gallbladder, connected to pancreas (beneath stomach) and duodenum of small intestine. Food travels to stomach (connected to duodenum). As food arrives at the stomach, it stretches the stomach walls, triggering release of the hormone gastrin. Gastrin circulates via the bloodstream back to the stomach, where it stimulates production of gastric juices (stimulation).
2. Chyme, an acidic mixture of partially digested food, eventually passes from the stomach to the duodenum. The duodenum responds to amino acids or fatty acids in the chyme by releasing the digestive hormones cholecystokinin and secretin. Cholecystokinin (CCK) stimulates the release of digestive enzymes from the pancreas and of bile from the gall bladder (stimulation). Secretin stimulates the pancreas to release bicarbonate (H C O 3 minus), which neutralizes chyme (stimulation).
3. If the chyme is rich in fats, the high levels of secretin and CCK released act on the stomach to inhibit peristalsis and secretion of gastric juices, thereby slowing digestion (inhibition).

Regulation of Energy Storage

When an animal takes in more energy-rich molecules than it needs for metabolism and activity, it stores the excess energy (see Concept 40.4). In humans, energy can be stored in liver and muscle as glycogen, a polysaccharide made up of many glucose units (see Figure 5.6b), or in adipose cells as triglycerides (fat). Glycogen stores are limited, but fat stores are not. A large grizzly bear can easily gain 75kg (165 lb) of fat as it prepares for hibernation.

At times when fewer calories are taken in than are expended—perhaps because of sustained heavy exercise or lack of food—muscle cells draw on their stored glycogen, and liver and adipose cells release glucose and fatty acids into the blood for use by other body cells. Fats are especially rich in energy; oxidizing a gram of fat liberates about twice the energy liberated from a gram of carbohydrate or protein. For this reason, *********adipose tissue** provides the most space-efficient way for the body to store large amounts of energy. Most healthy people have enough stored fat to sustain them through several weeks without food.

Glucose Homeostasis

The synthesis and breakdown of glycogen are central not only to energy storage, but also to maintaining metabolic balance through glucose homeostasis. In humans, the normal range for the concentration of glucose in the blood is 70–110 mg/100 mL. Because glucose is a major fuel for cellular respiration and a key source of carbon skeletons for biosynthesis, maintaining blood glucose concentrations near this set point is essential.

Glucose homeostasis relies predominantly on the antagonistic (opposing) effects of two hormones, insulin and glucagon **(Figure 41.23)**. When the blood glucose level rises above the set point, increased secretion of **insulin** triggers the uptake of glucose from the blood into body cells, decreasing the blood glucose concentration. When the blood glucose level drops below the set point, increased secretion of **glucagon** promotes the release of glucose into the blood from energy stores, such as liver glycogen, increasing the blood glucose concentration.

Figure 41.23

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Homeostatic regulation of cellular fuel.
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Flow diagram depicts the roles of hormones insulin and glucagon (both produced in the pancreas) in regulating blood glucose levels to normal (70 to 110 milligrams glucose per 100 milliliters).

After a meal is digested, glucose and other monomers are absorbed into the blood from the digestive tract. The human body regulates the use and storage of glucose, a major cellular fuel.

Flow diagram depicts the roles of the hormones insulin and glucagon in regulating blood glucose levels. Normal blood glucose is 70 to 110 milligrams glucose per 100 milliliters. Blood glucose level rises (such as after eating), and the beta cells of the pancreas secrete the hormone insulin into the blood. Insulin enhances the transport of glucose into body cells and stimulates the liver to store glucose as glycogen, resulting in the fall of blood glucose levels back to normal. From normal, blood glucose level can fall (such as after fasting), resulting in the alpha cells of the pancreas secreting the hormone glucagon into the blood. Glucagon promotes the breakdown of glycogen in the liver and the release of glucose into the blood, which results in blood glucose level rising back to normal.

MAKE CONNECTIONS *What form of feedback control do these regulatory circuits reflect (see*Concept 40.2*)?**

*For suggested answer, see*Appendix A.**

BioFlix® Animation

Homeostasis: Regulating Blood Sugar

image

Play

image

Animation

Pancreatic Hormones Regulate Blood Glucose Level

image

image

The liver is a key site of action for insulin and glucagon. After a carbohydrate-rich meal, for example, insulin promotes biosynthesis of glycogen from glucose that enters the liver in the hepatic portal vein. Between meals, when the blood glucose concentration decreases, glucagon stimulates the liver to break down glycogen, convert amino acids and glycerol to glucose, and release glucose into the blood. Together, these opposing effects of insulin and glucagon ensure that arterial blood has a glucose concentration within the set point at nearly all times.

Insulin also acts on nearly all body cells to stimulate glucose uptake from blood. A major exception is brain cells, which can take up glucose whether or not insulin is present. This evolutionary adaptation ensures that the brain almost always has access to circulating fuel, even if supplies are low.

Glucagon and insulin are both produced in the pancreas. Clusters of endocrine cells called pancreatic islets are scattered throughout this organ. Each islet has *alpha cells*, which make glucagon, and *beta cells*, which make insulin. Like other hormones, insulin and glucagon enter the interstitial fluid and then the circulatory system.

Overall, hormone-secreting cells make up only 1–2% of the mass of the pancreas. Other cells in the pancreas produce and secrete bicarbonate ions and the digestive enzymes active in the small intestine (see Figure 41.11). These secretions are released into small ducts that empty into the pancreatic duct, which leads to the small intestine. Thus, the pancreas has functions in both the endocrine and digestive systems.

Diabetes Mellitus

In discussing the role of insulin and glucagon in glucose homeostasis, we have focused exclusively on a healthy metabolic state. However, a number of disorders can disrupt glucose homeostasis with potentially serious consequences, especially for the heart, blood vessels, eyes, and kidneys. The best known and most prevalent of these disorders is diabetes mellitus.

The disease **diabetes mellitus** is caused by a deficiency of insulin or a decreased response to insulin in target tissues. The blood glucose level rises, but cells are unable to take up enough glucose to meet metabolic needs. Instead, fat becomes the main substrate for cellular respiration. In severe cases, acidic metabolites formed during fat breakdown accumulate in the blood, threatening life by lowering blood pH and depleting sodium and potassium ions from the body.

In people with diabetes mellitus, the level of glucose in the blood may exceed the capacity of the kidneys to reabsorb this nutrient. Glucose that remains in the kidney filtrate is excreted. For this reason, the presence of sugar in urine is one test for this disorder. The presence of excess sugar in both urine and blood is the basis for the name *diabetes mellitus* (from the Greek *diabainein*, to pass through, and *meli*, honey).

There are two main types of diabetes mellitus: type 1 and type 2. Each is marked by high blood glucose levels, but with very different causes.

Type 1 Diabetes

Also called insulin-dependent diabetes, *type 1 diabetes* is an autoimmune disorder in which the immune system destroys the beta cells of the pancreas. Type 1 diabetes, which usually appears during childhood, destroys the person's ability to produce insulin. Treatment consists of insulin injections, typically given multiple times daily. In the past, insulin was extracted from animal pancreases, but now human insulin can be readily obtained at reasonable cost from genetically engineered bacteria (see Figure 20.4). Stem cell research may someday provide a cure for type 1 diabetes by generating replacement beta cells that restore insulin production by the pancreas.

Type 2 Diabetes

Non-insulin-dependent diabetes, or *type 2 diabetes*, is characterized by a failure of target cells to respond normally to insulin. Insulin is produced, but target cells fail to take up glucose from the blood, and the blood glucose level remains elevated. Although heredity can play a role in type 2 diabetes, excess body weight and lack of exercise significantly increase the risk of developing this disorder. This form of diabetes generally appears after age 40, but even children can develop the disease, particularly if they are overweight and sedentary. More than 90% of people with diabetes have type 2. Many can control their blood glucose levels with regular exercise and a healthy diet; some require medications. Nevertheless, type 2 diabetes is a leading cause of death in the United States and a growing public health problem worldwide.

The resistance to insulin signaling in type 2 diabetes is sometimes due to a genetic defect in the insulin receptor or the insulin response pathway. In many cases, however, events in target cells suppress activity of an otherwise functional response pathway. One source of this suppression appears to be inflammatory signals generated by the innate immune system (see Concept 43.1). How obesity and inactivity relate to this suppression is being studied in both humans and laboratory animals.

BioFlix® Animation

Diabetes

image

Play

image

Regulation of Appetite and Consumption

Consuming more calories than the body needs for normal metabolism, or *overnourishment*, can lead to obesity, the excessive accumulation of fat. Obesity, in turn, contributes to a number of health problems, including type 2 diabetes, cancer of the colon and breast, and cardiovascular disease that can result in heart attacks and strokes. During the COVID-19 pandemic, obesity was identified as a major risk factor for COVID-19-associated hospitalization and death, particularly during the early waves of infection.

Researchers have discovered several homeostatic mechanisms that operate as feedback circuits controlling the storage and metabolism of fat. A network of neurons relays and integrates information from the digestive system to regulate secretion of hormones that regulate long-term and short-term appetite. The target for these hormones is a "satiety center" in the brain **(Figure 41.24)**. For example, *ghrelin*, a hormone secreted by the stomach wall, triggers feelings of hunger before meals. In contrast, both insulin and PYY, a hormone secreted by the small intestine after meals, suppress appetite. *Leptin*, a hormone produced by adipose (fat) tissue, also suppresses appetite and appears to play a major role in regulating body fat levels. In the **Scientific Skills Exercise**, you'll interpret data from an experiment studying genes that affect leptin production and function in mice.

Figure 41.24

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A few of the appetite-regulating hormones.
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Some hormones that act as appetite stimulants or suppressors described, including the hormones ghrelin, insulin, leptin and PYY.

Secreted by various organs and tissues, the hormones reach the brain via the bloodstream. These signals act on a region of the brain that in turn controls the "satiety center," which generates the nervous impulses that make us feel either hungry or satiated ("full"). The hormone ghrelin is an appetite stimulant; the other three hormones shown here are appetite suppressants.

Appetite Stimulant: Secreted by the stomach wall, ghrelin is one of the signals that act on the satiety center in the brain to trigger feelings of hunger as mealtimes approach. In dieters who lose weight, ghrelin levels increase, which may be one reason it's so hard to stay on a diet.

Appetite suppressants: 1. A rise in blood sugar level after a meal stimulates the pancreas to secrete insulin. In addition to its other functions, insulin suppresses appetite by acting on the satiety center in the brain. 2. Produced by adipose (fat) tissue, leptin suppresses appetite. When the amount of body fat decreases, leptin levels falls, and appetite increases. 3. The hormone PYY, secreted by the small intestine after meals, acts as an appetite suppressant that counters the appetite stimulant ghrelin.

Scientific Skills Exercise

Interpreting Data from an Experiment with Genetic Mutants

AP Science Practices 2, 3, 4, 5

What Are the Roles of theob**and**db**Genes in Appetite Regulation?**A mutation that disrupts a physiological process is often used to study the normal function of the mutated gene. Ideally, researchers use a standard set of conditions and compare animals that differ genetically only in whether a particular gene is mutant (nonfunctional) or wild-type (normal). In this way, a difference in phenotype, the physiological property being measured, can be attributed to a difference in genotype, the presence or absence of the mutation. To study the role of specific genes in regulating appetite, researchers used laboratory animals with known mutations in those genes.

Mice in which recessive mutations inactivate both copies of either the ob gene or the db gene eat voraciously and grow much more massive than wild-type mice.

Wild type thin mouse beside obese mouse with mutations to the o b gene.

The mouse on the right is wild-type, whereas the obese mouse on the left has an inactivating mutation in both copies of the ob gene.

One hypothesis for the normal role of the ob and db genes is that they participate in a hormone pathway that suppresses appetite when caloric intake is sufficient. Before setting out to isolate the potential hormone, researchers explored this hypothesis genetically.

How the Experiment Was Done The researchers measured the mass of young mice of various genotypes (the “subjects”) and surgically linked the circulatory system of each one to that of another mouse. This procedure ensured that any factor circulating in the bloodstream of either mouse would be transferred to the other in the pair. After eight weeks, they again measured the mass of each subject mouse.

Data from the Experiment

A table lists genotype pairing and the average change in body mass of the subject.

blank	Genotype pairing, subject	Genotype pairing, paired with	Average change in body mass of subject in grams
a.	o b + 1 a b +; d b + 1 d b +	o b + 1 o b +; d b + 1 d b +	8.3
b	o b 1 o b (mutant); d b 1 d b +	o b 1 o b (mutant); d b + 1 d b +	38.7
c.	o b 1 o b (mutant); d b + 1 d b +	o b + 1 o b +; d b + 1 d b +	8.2
d.	o b 1 o b (mutant); d b + 1 d b +	o b + 1 o b +; d b 1 d b (mutant)	negative 14.9, see note

Note, due to pronounced weight loss and weakening, subjects in this pairing were remeasured after less than 8 week.

Data from D. L. Coleman, Effects of parabiosis of obese mice with diabetes and normal mice, *Diabetologia* 9:294–298 (1973).

INTERPRET THE DATA

- First, practice reading the genotype information given in the data table. For example, pairing (a) joined two mice that each had the wild-type version of both genes. Describe the two mice in pairing (b), pairing (c), and pairing (d). Explain how each pairing contributed to the experimental design.
- Compare the results observed for pairing (a) and pairing (b) in terms of phenotype. If the results had been identical for these two pairings, what would that outcome have implied about the experimental design?
- Compare the results observed for pairing (c) to those observed for pairing (b). Based on these results, does the ob+ gene product appear to promote or suppress appetite? Explain your answer.
- Describe the results observed for pairing (d). Note how these results differ from those for pairing (b). Suggest a hypothesis to explain this difference. How could you test your hypothesis using the kinds of mice in this study?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology.**

Obtaining food, digesting it, and absorbing nutrients are part of the larger story of how animals fuel their activities. Provisioning the body also involves circulating nutrients, and using nutrients for metabolism requires exchanging respiratory gases with the environment. These processes and the adaptations that facilitate such distribution and exchange are the focus of Chapter 42.